Java 10 has Arrived! Check Out the New Features

Java 10 was released just over a few months ago and experts are still analysing its salient features. It has come out on schedule, true to the promise of a six-month release cycle that Oracle Corporation made after the release of Java 9. Here's a quick look at its features.

Hurray! Java 10 is out! Most of us are still not done with exploring Java 9, which was released on September 21, 2017, and exactly six months later, Java 10 has been released by Oracle Corporation. So, what is Java 10 all about? Let's find out.

An introduction to Java 10

JDK 10, i.e., the Java Development Kit, version 10, is an implementation of the Java SE 10 platform. This release was tracked by JEP 2.0 (Java Enhancement Proposal), which contains features to be developed in the specified release. Java 10 is available to the public and can be downloaded from Oracle's website. This release of Java is related to JSR 383 (Java Specification Request).

Java 10 was released on March 20, 2018, for general availability. It provides many new features, including local variable types, experimental features, etc. Java 10 was created in close collaboration with the OpenJDK Community — the collaborator for the open source implementation of the Java SE platform.

What are these new features?

Given below are the main new features introduced in Java 10. Let us discuss them in detail.

1. **Local-variable type inference:** Java 10 has introduced a new way of declaring and initialising local variables with the keyword 'var'. So, it uses:

```
var list = new ArrayList<String>();  // infers
ArrayList<String>
```

…instead of:

```
List<String> list = new ArrayList<String>();
```

This is restricted to local variables, which are indexes declared in loops. This feature is not available for method, constructor, method return types, fields, catch blocks, or any other kind of variable declaration. The main objective behind introducing this new data type is to enhance the Java language to extend type inference to declarations of local variables.

2. **Garbage Collector interface and parallel full GC for G1:** The Garbage Collector interface has been introduced to improve the source code isolation of different garbage collectors (GCs), hence providing better modularity for